



KCAC Position Statement: Non-Swallowed Formulations of Kratom Products

Kratom products that are formulated to not be swallowed should not be sold. This includes sublingual, oral disintegrating, buccal, vaping, inhaled, nasally inhaled (insufflated), transdermal, or injectable formulations.

Rationale

When a product is swallowed, the active ingredient(s) may not be fully absorbed. More importantly, the active ingredient(s) absorbed then go to the liver to be metabolized before reaching the blood stream ([first pass metabolism](#)). This is a standard biological safety mechanism to detoxify substances before they reach the bloodstream.

When products are formulated to bypass the liver, higher blood concentrations of the active ingredients can result, increasing the risk of adverse events. For this reason, the [Dietary Supplement Health and Education \(DSHEA\) Act of 1994](#) specifies that dietary supplements must be formulated to be swallowed.

A [recent assessment](#) conducted at the University of Connecticut School of Pharmacy revealed that almost 50 kratom derived products formulated to not be swallowed were being sold to U.S. consumers. These included sublingual (absorbed under tongue), buccal (absorbed by cheek and gum), and vaping (inhaled) formulations. A vast majority of these products contained concentrated synthetic 7-hydroxymitragynine (7-OH) or mitragynine pseudoindoxyl active ingredients, but some of the products contained mitragynine extracts.

The KCAC has [previously stated](#), consistent with [Food and Drug Administration \(FDA\) guidance](#), that concentrated synthetic 7-OH products should be designated Schedule I substances. Other non-swallowed formulations include inhaled, intranasal, transdermal, and injectable routes. There are [dietary supplement products](#) being sold using some of these formulations, but to our knowledge, no kratom-derived products as of yet.

Acceptable formulations include powders, capsules, tablets, and liquids. If products adhere to good manufacturing practices, are properly labeled, and include dosing aids that clearly reflect the recommended serving size, they are consistent with our [other position statements](#).